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heliostack: A Novel Approach to Minor Planet Discovery

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Abstract

The study of faint solar system objects is a promising avenue for understanding the origin and evolution of planetary systems. However, such objects are difficult to detect in conventional surveys. Here we introduce `heliostack`, an algorithm for nonlinear shift-and-stack searches for solar system objects, which enables us to combine images taken over longer time spans than was previously possible. Applying this algorithm to a number of existing archival and forthcoming surveys will allow us to maximize their potential for discovering faint solar system objects. In this work, we apply `heliostack` to archival Hubble Space Telescope (HST) data, completing an exhaustive search for cold classical Kuiper Belt objects in a set of HST images taken over a 15 day time span in 2003. We successfully recover both of the known subthreshold objects in the data, and add two new discoveries. These two new objects are the first to ever be discovered in stacks of images taken over a time span longer than about 1 day.

Unified Astronomy Thesaurus concepts: Solar system (1528); Kuiper belt (893); Planetary science (1255)

1. Introduction

Our solar system's small body populations offer a unique perspective for studying planetary systems. Today the census of minor planets stands at more than one million, which when taken in aggregate, reveal rich and delicate dynamical structures that have been sculpted over billions of years by the gravity of the planets. By studying the dynamical and physical properties of our solar system's populations of near-Earth objects (NEOs), asteroids, Jupiter Trojans, Centaurs, comets, Neptune Trojans, Kuiper Belt objects (KBOs), and beyond, we have gained deep insight about the processes driving the formation and evolution of planetary systems. Still, fundamental questions remain: Is there a massive planet still awaiting discovery in the distant solar system (K. Batygin et al. 2019)? Are there more populations of small bodies that have thus far eluded detection? To answer these questions we must develop techniques to discover fainter objects so that we can probe smaller and more distant populations.

In the two centuries since scientists began working to detect and characterize solar system bodies, progressive improvements in telescope and detector technology have greatly enhanced our capabilities. However, we are reaching the physical limits of detectors and the practical engineering limits of telescopes, and since the objects we are searching for are moving, we cannot simply take longer exposures to increase image depth. One way that we can continue to make progress without improving hardware, is by developing software and algorithms that can better utilize our data.

There is a standard procedure for detecting moving objects fainter than the single-image detection threshold, called shift and stack, in which one stacks images along the trajectory of some putative object (J. A. Tyson et al. 1992; B. Gladman et al. 1998; R. L. Allen et al. 2002; G. M. Bernstein et al. 2004; M. J. Holman et al. 2004; H. Smotherman et al. 2021; K. J. Napier et al. 2024). When the signal from the object is combined in a sufficient number of images, the object becomes detectable. The problem, however, is that we do not usually have a priori knowledge of the trajectories of the real objects in our images. In those cases, we must search through all possible orbits of interest. Historically, shift-and-stack searches for solar system objects have not explored a full six-dimensional orbital parameter space, but have instead remained limited to a four-dimensional regime where an object's apparent motion remains linear in pixel space. In this regime, it is simple to determine which trajectories need to be checked for an exhaustive search (see K. J. Napier et al. 2024, for example). However, the requirement for linear trajectories limits stacks to data taken over the span of a few hours. When the duration of the exposure sequence gets any longer, the apparent motion of the objects becomes nonlinear in pixel space, increasing the complexity of the problem. Because of the increased complexity over longer time spans, the shift-and-stack technique has thus far been restricted to dedicated shift-and-stack surveys, which typically take of order 100 images of the same patch of sky in a given night (D. E. Trilling et al. 2024; C. A. Trujillo et al. 2024)

To date, only G. M. Bernstein et al. (2004) have completed a systematic shift-and-stack search for solar system objects in images that were taken over a time span longer than a few hours. There has been recent work in the area of nonlinear digital tracking (see, e.g., N. Golovich et al. 2025), but so far no systematic searches have been completed, and no new objects have been discovered. In the 2 decades since G. M. Bernstein et al. (2004), with the emergence of GPUs and incredible improvements in computing capacity on CPU clusters, why have not any other shift-and-stack searches over long time baselines been completed? The answer may simply

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be that the problem is difficult from both theoretical and computational perspectives. Just enumerating the trajectories to be searched requires a careful choice of parameters, and even with an efficient parameterization the number of trajectories to search can quickly become unmanageable.

Despite its immense difficulties, the problem is worth solving. If we can manage to combine images taken days, months, or even years apart, we will be able to reach the depth of a shift-and-stack search using data from surveys with sparser cadences such as the Dark Energy Survey (DES), or the Legacy Survey of Space and Time (LSST). If we can gain 1 or 2 mag of sensitivity over such large survey areas, we will see a substantial increase in the yield of solar system objects, with especially important gains for populations of objects that are small or very distant (M. Jurić et al. 2019; C. A. Trujillo et al. 2024). Furthermore, if we can manage to search for solar system objects in sparser or more irregular image cadences, we can develop much more flexible survey plans, making it easier for the community of solar system observers to obtain time on the most oversubscribed telescopes such as the James Webb Space Telescope (JWST) or the Nancy Grace Roman Space Telescope (Roman).

In this paper we introduce `heliostack`, an algorithm designed to enable nonlinear shift-and-stack searches for solar system objects. One of the key insights that enables `heliostack` to efficiently search through the space of nonlinear trajectories is inspired by the eponymous `HelioLinC` algorithm (M. J. Holman et al. 2018). Given detections of some solar system object, guesses of the object's unmeasured orbital parameters can be used to fully define an orbit for each detection. If the guess was correct (or at least close enough), those orbits will cluster on the sky when propagated to a common epoch. In retrospect, the idea is obvious—no object can be in two places at once! We take this idea a step further in `heliostack`, applying the idea not just to catalogs of tracklets, but rather treating each pixel in each image as a "detection," and using some set of asserted orbital parameters to synchronize our pixels to a common epoch.

We begin by describing the `heliostack` algorithm in Section 2. In Section 3, we demonstrate the algorithm by searching for KBOs in the Hubble Space Telescope (HST) data from G. M. Bernstein et al. (2004). Finally, in Section 4, we identify some immediate applications of the `heliostack` algorithm, and speculate as to how it may be useful in the future, as both data volume and computing capacity continue to grow.

2. The Algorithm

We operate under the hypothesis that we have some images that contain an object that is too faint to be confidently detected in individual exposures. Our goal is to either find this object or, in failing to do so, place some upper limit on how bright it may be. In order to do this, we need to have prior knowledge of the object's location in each of our images, but fortunately we do not need to have prior knowledge of its brightness. While objects' brightnesses can be variable in complicated ways, their trajectories are usually deterministic from a given initial condition, and are therefore simple to describe.[6] In particular, the motion of solar system objects can be described by some appropriate set of six orbital parameters at a given time. But how can we choose our parameters to completely, and nonredundantly, trace out the paths that any possible moving object may take on the sky as seen by some moving observatory? The problem is complicated, and becomes even more complicated if one considers observations from multiple observatories. We can simplify the problem by using an approach inspired by `HelioLinC`.

Let us begin with the idea of generating a blank canvas of the barycentric sky at some reference epoch, $t_{\rm ref}$. Locations on this canvas can be described by unit vectors with an origin at the barycenter. We will use this canvas to accumulate signal from our images, so it should be made up of discrete bins (preferably pixels of approximately the same angular size as the pixels in the images that are being combined). By stacking images onto this canvas, we can decouple any topocentric observer information. But how do we determine where on our canvas each pixel from each image should go?

First, we can recognize that a pixel in an image is associated with a topocentric sky position $\{\alpha, \delta\}$, a time t, a flux, a flux uncertainty, and a vector describing the observer's position relative to some inertial frame, $\boldsymbol{x}_{\rm obs}$. The topocentric sky position of the pixel can be expressed as a Cartesian unit vector:

$$\hat{\boldsymbol{\rho}} = \langle \cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta \rangle. \tag{1}$$

The barycentric position vector to the object at the time when it reflected the observed photons can be expressed as

$$\boldsymbol{r}\left(t - \frac{\rho}{c}\right) = \boldsymbol{x}_{\rm obs} + \rho\hat{\rho}. \tag{2}$$

where ρ is the topocentric distance to the object, and ρ/c is the light time.

The direction of the object's barycentric unit vector at the reference epoch dictates where it should be registered to on the canvas. In order to properly register all pixels to the canvas, then, we need to know $\boldsymbol{r}(t)$, which requires knowing all six orbital parameters. Notice that two of the required six pieces of information can come directly from the image itself, α and δ. In order to fully specify the orbit, then, we need to assert four more orbital parameters.

The first three orbital parameters we choose to assert are the barycentric distance r, the line of sight velocity v_r, and the tangential velocity, v_Ω, which are components of a basis that is described in detail in K. J. Napier & M. J. Holman (2024). Under the assumption of two-body motion, the universal anomaly, $\chi(t)$, can be calculated at any time t as long as we know r and v_r at a reference epoch t_0, the specific energy, ε, and the time interval, $t - t_0$ (J. M. A. Danby 1992). The expression for the position vector as a function of time can be written as

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 f + \boldsymbol{v}_0 g, \tag{3}$$

where

$$f(t) = 1 - \frac{\chi(t)^2}{r_0} C(z), \tag{4}$$

and

$$g(t) = (t - t_0) - \frac{\chi(t)^3}{\sqrt{\mu}} S(z). \tag{5}$$

[6] We consider only Keplerian orbits in the present treatment.

Here $z = \chi(t)^2/a = -2\varepsilon\chi(t)^2/\mu$, and $C(z)$ and $S(z)$ are the Stumpff functions (J. M. A. Danby 1992). We can take the vector dot product of both sides of Equation (3) to yield an equation for the scalar barycentric distance, r, at any given time:

$$r(t)^2 = r_0^2 f(t)^2 + v_0^2 g(t)^2 + 2f(t)g(t)r_0 v_{r,0}. \tag{6}$$

The fourth parameter to assert is the orbital inclination, i, as well as a value called $\kappa \in \{+1, -1\}$, which breaks the degeneracy between ascending and descending orbits at the epoch of the elements (see K. J. Napier & M. J. Holman 2024). Our asserted $\{r, v_r, v_\Omega, i\}$ map onto the orbital invariants. An orbit's specific energy, ε, specific angular momentum, h, and the z-component of the angular momentum, h_z, can be expressed as

$$\varepsilon = \frac{v_r^2 + v_\Omega^2}{2} - \frac{\mu}{r}, \quad h = r v_\Omega, \quad \text{and } h_z = r v_\Omega \cos i. \tag{7}$$

The mapping of our asserted orbital parameters onto the Keplerian invariants is an extremely useful property because it means that we can simultaneously solve for the orbits of all pixels from all images, making the problem easy to parallelize.

There is at most one orbit that has the observed or known quantities $\{\alpha, \delta, \boldsymbol{x}_{\rm obs}, t\}$, and the asserted quantities $\{r, v_r, v_\Omega, i, \kappa, t_{\rm ref}\}$, which we can solve for as follows.[7] First we can take the vector dot product of Equation (2) with itself to put it into scalar form:

$$\left[r\left(t - \frac{\rho}{c}\right)\right]^2 = |\boldsymbol{x}_{\rm obs}|^2 + \rho^2 + 2\rho(\hat{\rho} \cdot \boldsymbol{x}_{\rm obs}). \tag{8}$$

It is possible to find a value of ρ that solves Equation (8) because as previously established, $r(t)$ is uniquely determined at all times.[8] Plugging our solution for ρ back into Equation (2), we can get the position vector to the orbit at the epoch when the object reflected the observed photons. To completely specify the orbit, we also need to find its velocity vector at the light-corrected epoch. We can do so by using the approach described by K. J. Napier & M. J. Holman (2024), who show that we can express the state vector as

$$\boldsymbol{r} = r\hat{r}, \tag{9}$$

$$\boldsymbol{v} = v v_r \hat{r} + v_\Omega(\cos\psi \hat{A} + \sin\psi \hat{D}), \tag{10}$$

where we have now introduced ψ, which describes the direction that v_Ω points in the plane defined by $\hat{A}$ and $\hat{D}$, which are given by

$$\hat{A} = \langle -\sin\varphi, \cos\varphi, 0\rangle, \tag{11}$$

$$\hat{D} = \langle -\sin\theta\cos\varphi, -\sin\theta\sin\varphi, \cos\theta\rangle. \tag{12}$$

We already have $\{r, \varphi, \theta\}$ at the light-corrected epoch from Equation (2), so the remaining undetermined parameters are v_r, v_Ω, and ψ. Since we have specified the orbital invariants, we can solve for those remaining parameters as follows:

$$v_\Omega = \frac{h}{r}, \tag{13}$$

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} \;\Rightarrow\; v^2 = 2\left(\epsilon + \frac{\mu}{r}\right), \tag{14}$$

$$v_r = \sqrt{v^2 - v_\Omega^2}, \tag{15}$$

$$\cos\psi = \frac{\cos i}{\cos\theta}, \tag{16}$$

$$\sin\psi = \kappa\sqrt{1 - \cos^2\psi}. \tag{17}$$

Now the orbit's position and velocity at the light-corrected epoch are fully specified, so it can be propagated to the reference epoch of the initial condition and we can accumulate the signal onto the canvas accordingly. Each pixel in the canvas corresponds to an orbit which has orbital parameters given by the pixel's φ and θ, and the initial condition's r, v_r, v_Ω, ψ, and $t_{\rm ref}$. Any pixel with sufficiently high significance can be thought of as a candidate moving object, which can later be investigated more carefully.

3. Demonstration on Archival HST Data

As a proof of concept for the `heliostack` algorithm, we have reprocessed the data from G. M. Bernstein et al. (2004) in a targeted search for cold classical KBOs (CCKBOs). The images from this dataset were obtained in 2003 during 125 HST orbits over a time span of about 15 days, and cover about 0.02 deg^2 with six dithered, nearly contiguous Advanced Camera for Surveys (ACS) Wide Field Channel (WFC) fields of view near the solar system's invariable plane. Each of the survey's pointings was imaged for a total integration time of about 38,000 s, broken into 95 exposures of approximately 400 s each. These images were strategically taken near quadrature, so that the trans-Neptunian objects (TNOs) in the field would be near their semiannual parallax turnaround point at the midpoint of the survey, preventing them from drifting out of the survey's extremely small field of view. More detailed information about the images can be found in G. M. Bernstein et al. (2004). We used the MAST archive to acquire the `.flc` images of this field, which were processed by HST's `CALACS` pipeline, and thus were calibrated, charge transfer efficiency corrected, and given internally consistent astrometric solutions.

When G. M. Bernstein et al. (2004) processed these data more than 20 yr ago, it was computationally prohibitive to stack images taken over a span of more than 24 hr. To compensate for this limitation, they stacked chunks of the data, and then linked the stacked detections across multiple epochs. In this work, since it is now computationally feasible, we simply stack the entire dataset.

3.1. Image Preprocessing

We had to take a few simple steps to prepare our images to be searched. At each step, we treated both our images and their corresponding variance so that we could properly weight our images in the stacks. We began by injecting a synthetic population of CCKBOs for later use in quantifying our detection efficiency. It is best for this step to occur before any further processing, so that the synthetic sources undergo the

[7] If a pixel's latitude exceeds its putative inclination, no object with the chosen orbital parameters could possibly be on that pixel, and so we can simply ignore it.

[8] Note that in the case of a Keplerian orbit, $r(t)$ is completely decoupled from the orbit's orientation in space. This property enables the precalculation of $r(t)$ at all times for all possible sky locations, thus eliminating a lot of redundant computation.

same processing as real sources to the extent possible. Our synthetic population had apparent magnitudes between 28.4 and 29.7 at the survey's midpoint, and single-component sinusoidal light curves with amplitudes as large as 1 mag, and periods between 4 and 36 hr. We implanted the flux using an empirical point-spread function (PSF) model (G. S. Anand et al. 2023) with appropriate trailing.

The rest of the steps can be enumerated as follows.

1. Convert all images to units of electrons per second.
2. Fit and subtract a spatially varying background from each image.
3. Mask regions with 500 or more contiguous pixels brighter than 5σ.
4. Resample each image onto a rectilinear grid with a uniform pixel scale of $0\farcs05$, using Lanczos-4 interpolation.
5. Generate a template for each CCD by aligning the images and calculating a robust Huber mean (P. J. Huber 1964).[9]
6. Create difference images by subtracting the corresponding template from each image.
7. Extract all sources brighter than 3.5σ from each image. These catalogs could be used for a search for bright objects, but we do not do a bright search in this analysis.
8. Flag and mask all pixels brighter than 4σ, as well as a 2 pixel buffer. This step is important for avoiding false positives in the stacks.

Since HST's PSF and seeing are so stable, the image subtraction worked well without any PSF matching or rescaling. After masking bright transients (i.e., cosmic rays, asteroids, and TNOs), we find that each of our images was well approximated as Gaussian noise without correlations. This was a critical detail for avoiding false positives in the stack.

Finally, we generated two new kinds of images, defined as follows:

$$\xi = \left(\frac{D}{V}\right) \star P, \tag{18}$$

and

$$\zeta = \left(\frac{1}{V}\right) \star P^2, \tag{19}$$

where D is the difference image, V is its variance, P is the image's PSF, and $\star$ represents a cross-correlation operation. These preconvolved ξ and ζ images are optimal for point-source detection, and make it trivial to calculate a weighted average with a PSF-matched filter when stacking (G. M. Bernstein et al. 2004; D. Lang & D. W. Hogg 2026). Notice that ζ is, by definition, just the variance of ξ, so the significance of any pixel is given by $\nu = \xi/\zeta^{0.5}$. One drawback to treating the images this way is that since we only convolve the images with their PSF one time, they are not quite optimal for the detection of trailed sources. This limitation is not a big issue though, because after the initial identification of candidates, it is easy to investigate them more carefully (see Section 3.4).

[9] We found that the robust mean was successful at preventing transients (including both cosmic rays and real moving solar system objects) from appearing in the template; our templates caused no measurable self-subtraction of our implanted synthetic objects.

3.2. Grid of Initial Conditions

The final bit of preparation was to choose which orbits to search. Since the space of orbital parameters is continuous there are, in principle, infinitely many possibilities. In practice, however, the search space consists of a finite set of initial conditions that guarantees tracking errors to be smaller than some tolerance ε for all points in the parameter space of interest. That is, no trajectory of interest will deviate from a searched trajectory by more than ε. For this search, we chose $\varepsilon = 0\farcs05$, comparable to the PSF width and pixel scale of ACS/WFC. Tracking errors at this level can, in principle, degrade the depth of our search by up to 0.1–0.2 mag, but stacking at a finer resolution, for instance $\varepsilon = 0\farcs03$ as done by G. M. Bernstein et al. (2004), would represent a fourfold increase in computing resources, which we consider to be unnecessary for this proof of concept.

For this demonstration, we generated a grid of initial conditions by selecting values of r, $v_\Omega \cos\psi$, and $v_\Omega \sin\psi$ that yield tracking errors smaller than $0\farcs05$ over the 15 day time baseline of our data (M. J. Holman et al. 2019). Over such short arcs, the radial velocities of the CCKBOs do not make any measurable difference to their sky motion, so we do not lose any sensitivity by using $v_r = 0$ for each initial condition. The problem becomes more complicated for longer arcs where deviations in v_r become detectable, but it is still solvable (see T. R. Ruch et al. 2026, in preparation, for one approach). We pare our initial conditions to the parameter space of the CCKBOs with bounds $a \in [42.4, 47.4]$ au, $e \in [0, 0.2]$, and $i_{\rm ecliptic} \in [0°, 5°]$. We also restrict $r \in [38, 51]$ au, as this range includes more than 95% of the CCKBOs (J. J. Kavelaars et al. 2021). These restrictions result in a total of 1,148,817 initial conditions to check. Checking the full Kuiper Belt beyond 38 au, with inclinations less than 60°, would require a factor of 300 more initial conditions (though it is likely that the grid could be made smaller by eliminating some redundant trajectories).

3.3. Stacking and Candidate Identification

With our images preprocessed and a list of orbits to check, we proceeded with the search. For each orbit, we began by generating a blank canvas with a linear world coordinate system (WCS) of barycentric (φ, θ) at a reference epoch of the survey's midpoint.[10] We then synchronized each image to our canvas using the transformation described in Section 2. In principle we could have done this synchronization pixel by pixel. However, since our images were reprojected to have a linear WCS in the preprocessing stage, we were able to instead calculate the transformation for each of the corners of an image, and use the transformed points to calculate the perspective transformation matrix to project the topocentric coordinates onto the barycentric canvas (see, e.g., R. Hartley & A. Zisserman 2004).[11] This approximation provides a significant speedup, especially on a GPU, because it takes advantage of the fact that images are spatially indexed and contiguous in memory. We then applied this perspective transformation to the image, and added the image to our

[10] The tangent plane is arbitrarily defined with respect to the lower left corner of the canvas in each stack.

[11] We find empirically that errors introduced by the perspective projection (as opposed to going pixel by pixel) remain smaller than 0.2 pixels over a full image, providing plenty of fidelity for initial candidate identification.

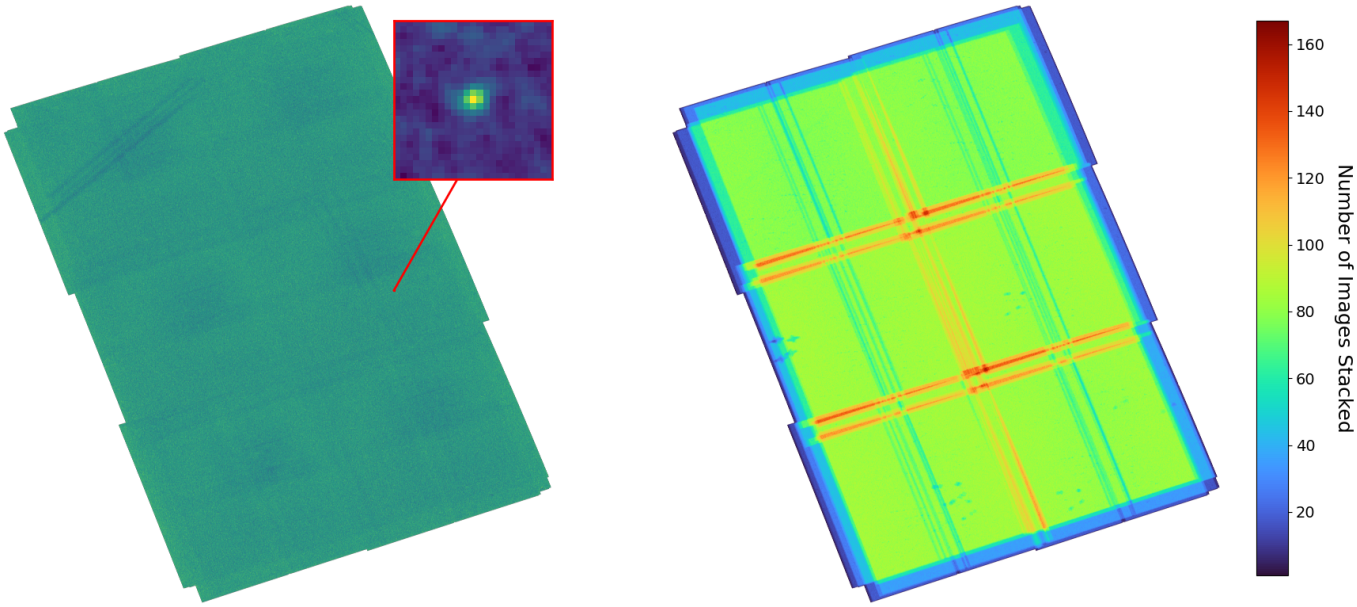


Figure 1. Left: the canvas (approximately 14,000 × 13,000 pixels) after stacking all 1118 images for some initial condition. The inset plot shows the lone peak above 7σ in the stack, which turns out to be a 21σ detection of 2003 BH_{91}. Since the asserted parameters are not quite correct, the detection is somewhat smeared. Right: a heat map of the number of images that contributed to each pixel in the significance image.

canvas accordingly, keeping running sums of both ξ and ζ. Since this stage of the search is very approximate, we used nearest neighbor interpolation to make the stacks as fast as possible. More complicated interpolation schemes would also work without much of a performance penalty, but the exact resampling scheme does not matter much at this stage in the process. After all of the images were synchronized to the canvas for a given orbit, we calculated a significance image, $\nu = \xi/\zeta^{0.5}$, ran a peak finding routine to find all local maxima on the canvas, and took note of all peaks with $\nu \geqslant 7.0$. For an example of the significance image, see the left-hand panel of Figure 1.

Searching the 1118 images with 1,148,817 initial conditions yielded a total of 109,383 candidate objects, each associated with a complete orbit specification, $\{\varphi, \theta, r, v_r, v_\Omega, \psi\}$ at $t_{\rm ref}$, as well as a flux and a flux uncertainty. Many of these candidates turned out to be redundant detections of the same object, but found with slightly different significances and with different asserted parameters. To eliminate such repeated detections, we identified objects whose trajectories coincided to within $0\farcs2$ in both R.A. and decl. in at least 30 images, and retained the candidate with the highest ν from each group of associated trajectories. This step resulted in a reduction to 533 total candidates with $\nu \geqslant 7.0$. Still, a few of the trajectories were redundant detections of the same object, but found along different segments of the object's trajectory; these duplicates are resolved after the orbit tune-up procedure described in Section 3.4. Of the 533 candidates, 165 would later be identified as implanted sources, meaning that there were 368 sources remaining that were not associated with an implanted synthetic source, which is roughly consistent with the expected number of sources with $\nu > 7$, assuming Gaussian noise and $\sim 2 \times 10^{14}$ trials (see the distribution of ν in the top panel of Figure 2). It is a testament to the quality of these data, and the importance of using clean difference images, that the candidate list was so manageably small without any machine learning or visual inspection for false positive rejection. At this point, each of our candidates was still just associated with a discrete orbit from our list of initial conditions, and needed to be tuned up in a continuous parameter space.

3.4. Candidate Tune-up

After finding our candidates in the discrete search, we optimized each source's significance by fitting a six-parameter orbit directly on the pixels. For this tune-up procedure we used a different stacking approach, working directly with the nonconvolved difference images. For each candidate, we chose a six-parameter orbit given by $\{\alpha, \delta, r, v_r, v_\Omega, i, \kappa\}$ at $t_{\rm ref}$, propagated the orbit to the epoch of each image, and used a trailed PSF model to obtain forced measurements of the flux and flux uncertainty of the putative source.[12] This process generated a light curve for each object, which we then used to calculate a sigma-clipped (3σ with 10 iterations) mean flux and flux uncertainty. Taking the ratio of the mean flux to the flux uncertainty yields a significance, ν, and we used $\nu^2/2$ (which is the Gaussian log likelihood) as an optimization metric for a downhill simplex optimizer. The result of this optimization procedure is shown in the bottom panel of Figure 2. In general, real candidates tend to increase in significance, while spurious candidates tend not to. There is a clear boundary at $\nu = 8.8$, beyond which there are likely no false positives. Note that one major difference between the top and bottom panels is that the ν values in the tune-up procedure were calculated using a robust (sigma-clipped) mean, while the initial candidates' significances were derived from inverse-variance-weighted means, with no outlier rejection; this is the reason why some candidates have $\nu < 7$ after tune-up.

[12] Using the topocentric observables helped to constrain the solutions, because the observer's position is fixed at the reference epoch, meaning that the solution for an observation's barycentric sky position (φ, θ) is coupled to its barycentric distance.

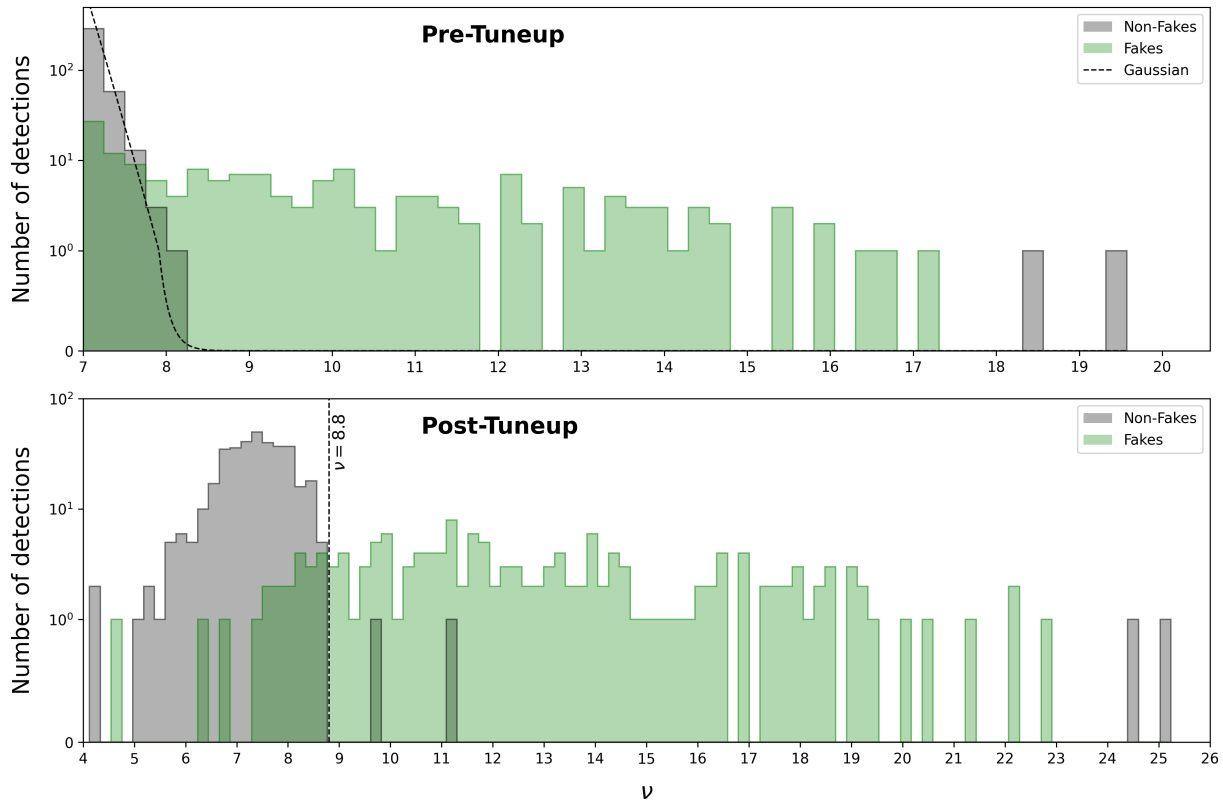


Figure 2. Top: distribution of the signal-to-noise ratio of all of the unique candidate trajectories resulting from the `heliostack` search, with $\nu \geqslant 7$. The green histogram represents the implanted synthetic sources, and the gray histogram represents real candidate moving objects. The dashed black line shows the shape of a Gaussian distribution with a mean of zero and a width of one. It is clear that the distribution of ν is very nearly Gaussian. Bottom: distribution of the signal-to-noise ratio of all of the unique candidate trajectories after the tune-up step described in Section 3.4. Real candidates tend to increase in significance, while spurious candidates tend not to. There is a clear boundary at $\nu = 8.8$, beyond which there are probably no false positives.

3.5. Results

After tune-up, we retained all candidates with $\nu \geqslant 8.8$, which is well motivated by Figure 2. Our final list contained 145 candidates, 141 of which turned out to be implanted sources. Our efficiency, shown in Figure 3, is reasonably well described by a function of the form

$$\eta(m) = \frac{\eta_0}{1 + \exp\left(\frac{m - m_{50}}{\sigma}\right)}, \tag{20}$$

with $\eta_0 = 1.0$, $m_{50} = 29.21 \pm 0.03$, and $\sigma = 0.11 \pm 0.02$. Note that we used a maximum likelihood approach to fit our efficiency, rather than binned data. One notable feature of our efficiency function is a rather long tail, which can be attributed to increased sensitivity in the regions where chips overlap in the stack (see the right-hand panel in Figure 1).

The remaining four candidates are, as far as we can tell, real objects. Two of the objects, 2003 BF_{91} and 2003 BH_{91}, were previously discovered by G. M. Bernstein et al. (2004). The other two objects, which we call 2003 ABCD and 2003 WXYZ as they do not yet have Minor Planet Center designations, are newly discovered in this search; they were likely missed in the search by G. M. Bernstein et al. (2004) because they were simply too faint to be found without stacking all of the survey's images simultaneously. We show cutouts of the objects in Figure 4, and list their photometric and orbital properties in Table 1. The cutouts of these objects are indistinguishable from the cutouts of the recovered synthetic sources, shown in Figure 5. The uncertainties in the orbital parameters come from a Markov Chain Monte Carlo (MCMC) simulation in which we allowed all six orbital parameters to vary, with uniform priors. Note that we did not bother searching for the two brighter objects in this dataset, 2000 FV_{53} and 2003 BG_{91}, as we consider finding objects that can be seen in individual exposures to be a solved problem for data with this survey's cadence.

While G. M. Bernstein et al. (2004) reported a limiting magnitude of $m_{50} = 29.17$, similar to our limiting magnitude of $m_{50} = 29.21 \pm 0.03$, we believe that our search actually went significantly deeper, and that there is some discrepancy with the magnitudes reported. The first piece of evidence comes from the measured magnitudes of the objects detected by both analyses. We measure the mean magnitude of 2003 BF_{91} to be 27.98 ± 0.04, while G. M. Bernstein et al. (2004) reported 28.15 ± 0.04. Similarly, we measure the mean magnitude of 2003 BH_{91} to be 28.16 ± 0.04, while G. M. Bernstein et al. (2004) reported 28.38 ± 0.05. Second, we have confident detections of both 2003 ABCD and 2003 WXYZ, while G. M. Bernstein et al. (2004) did not find either object.

One possible reason for the magnitude discrepancy is some difference with the zero-points used in the analyses. An erratum from G. M. Bernstein et al. (2004, 2006) may support this idea of mismatched zero-points, though it is difficult to know for certain. It is also worth noting that the images have undergone different processing by more recent versions of the `CALACS` pipeline since

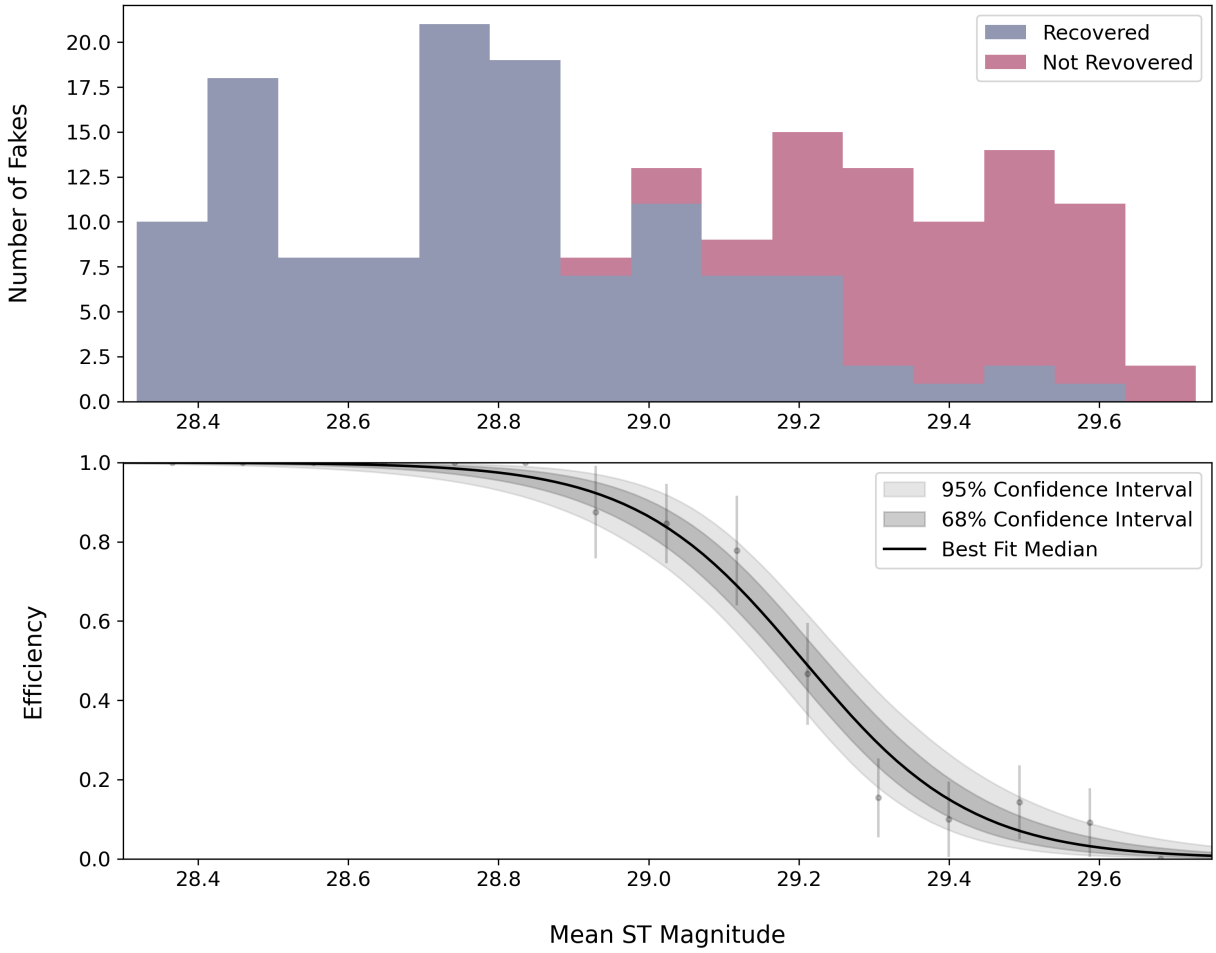


Figure 3. Top: histogram of the mean apparent magnitude of the implanted synthetic sources, with an accounting of whether or not the sources were recovered. Bottom: best fit of our efficiency using Equation (20), including 68% and 95% confidence intervals. The best-fit parameters are $\eta_0 = 1.0$, $m_{50} = 29.21 \pm 0.03$, and $\sigma = 0.11 \pm 0.02$.

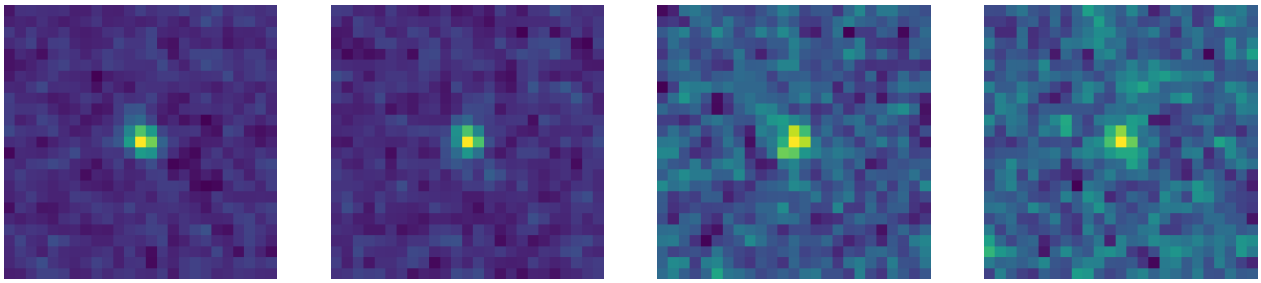


Figure 4. From left to right, stacked cutouts of 2003 BF_{91}, 2003 BH_{91}, 2003 ABCD, and 2003 WXYZ.

Table 1
Photometric and Orbital Parameters of the Detected Objects

Object ID	Flux (e^- s^{-1})	ν	Total Integration (s)	STMag	r_{bary} (au)	$i_{ecliptic}$ (deg)
2003 BF_{91}	0.299 ± 0.012	25.2	27,622	27.99 ± 0.04	$42.19^{+0.01}_{-0.01}$	$1.50^{+0.001}_{-0.001}$
2003 BH_{91}	0.255 ± 0.010	24.4	33,576	28.16 ± 0.04	$42.59^{+0.02}_{-0.02}$	$2.00^{+0.014}_{-0.013}$
2003 ABCD	0.119 ± 0.011	11.1	29,875	28.99 ± 0.10	$48.95^{+0.05}_{-0.05}$	$2.00^{+0.025}_{-0.028}$
2003 WXYZ	0.100 ± 0.010	9.7	32,458	29.17 ± 0.11	$42.73^{+0.06}_{-0.05}$	$3.04^{+0.015}_{-0.023}$

the search by G. M. Bernstein et al. (2004). Regardless of the cause for the discrepancy, it seems that our search went about 0.2 mag deeper. Since our stacks had an effective integration time of 38,000 s, compared to 22,000 s for the search by G. M. Bernstein et al. (2004), the theoretical maximum increase in depth was $1.25 \log_{10}(38,000/22,000) \approx 0.3$ mag.

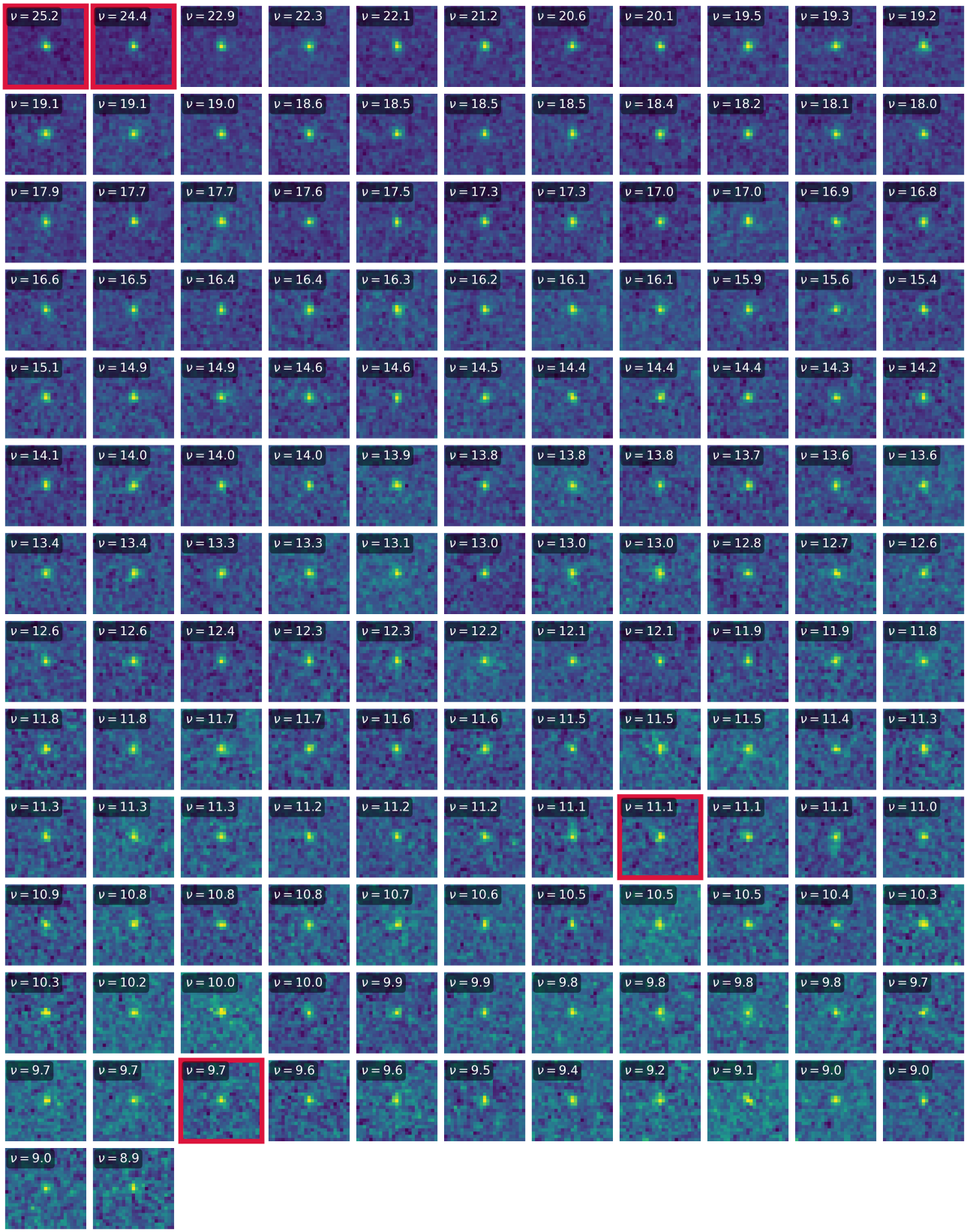


Figure 5. Cutouts of each of the recovered sources. Objects that are not implanted synthetic sources are highlighted in red.

Given that we used a significance cutoff of $\nu = 8.8$, rather than $\nu = 8.2$ as done by G. M. Bernstein et al. (2004), we would expect a loss of $-2.5\log_{10}(8.2/8.8) \approx 0.08$ mag, so our gain of about 0.2 mag rather than 0.3 mag is more or less expected.

Of the five KBOs in this dataset (excluding the targeted KBO 2000 FV_{53}), only 2003 BG_{91} can be securely classified as a CCKBO. The arcs are simply too short for the orbits to be well determined. The other objects, 2003 BF_{91}, 2003 BH_{91},

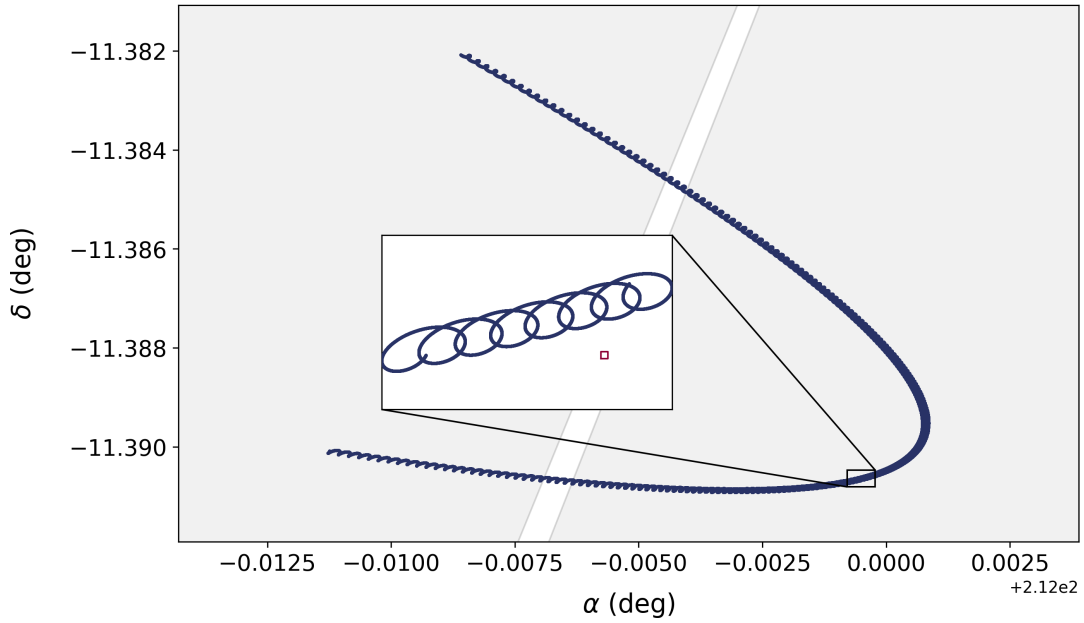


Figure 6. Trajectory of 2003 BF_{91} on the sky as seen by HST over the course of the survey. The inset plot emphasizes how nonlinear the trajectory is, due to HST's low-Earth orbit; the proper motion in the sky plane remains linear for at most a few minutes at a time. The red square shows the size of 1 pixel; it is clear that the parallax loops are too large to simply ignore. The `heliostack` algorithm automatically corrects for this parallax. The gray regions represent the locations of different CCDs, showing that we are able to find objects that cross detectors over the course of the survey.

Table 2
Derived Orbital Limits

Object ID	95% Confidence Interval for i_{free} Assuming Cold Classical	a_{min} (au)	e_{min}
2003 BF_{91}	1.°08–1.°62	42.7	0.014
2003 BH_{91}	0.°09–1.°01	43.9	0.031
2003 ABCD	0.°49–0.°53	47.5	0.026
2003 WXYZ	1.°61–1.°81	45.3	0.059

2003 ABCD, and 2003 WXYZ, may be CCKBOs given their measured barycentric distances and orbital inclinations, but cannot be definitively classified because their orbital elements are not well enough constrained by the 15 day arcs. For the MCMC samples of each object where their a and e values were consistent with cold classical orbits, we calculated the free inclinations as prescribed by Y. Huang et al. (2022). We report 95% confidence intervals for this experiment, along with the minimum allowed values of a and e, in Table 2. For each object, given the assumption that the a and e values are consistent with being CCKBOs, the free inclinations are also consistent with being CCKBOs. It is interesting to note, however, that 2003 ABCD has a minimum semimajor axis of $a \approx 47.5$ au, suggesting that it may not truly be a CCKBO, but rather may be in a 2:1 mean motion resonance with Neptune. If this is the case, the Hamiltonian model used to calculate its free inclination is incorrect, meaning that the result does not hold.

4. Discussion and Conclusions

In this paper we have introduced `heliostack` as a solution to the problem of nonlinear shift and stack, and applied it to archival HST data. We used the algorithm to exhaustively search for CCKBOs in a set of 1118 images taken over a 15 day time span, successfully recovering both of the known subthreshold objects in the data, and adding two new discoveries. We did not perform a search for bright objects in these data, so we did not report measurements for 2000 FV_{53} and 2003 BG_{91}, but we could see them in the images. As far as we are aware, 2003 ABCD and 2003 WXYZ are the first objects to be discovered in stacks of images with a time baseline longer than about 1 day. The algorithm did not have any problem finding objects that moved between chips, and was able to automatically correct for the extremely nonlinear on-sky trajectories of the objects caused by HST's orbit around Earth (see Figure 6).

The search process was extremely efficient; all of the steps discussed in this paper were run in a few hundred GPU hours on NVIDIA H200 cards, and a few thousand CPU hours. From start to finish, the process took less than 1 day on Harvard's Cannon cluster. Furthermore, in this case the production of the final catalog did not require any machine learning or visual inspection. A search of more than 10^{14} trajectories resulted in a list of just 145 candidates, where 141 of which were implanted sources, and other four are almost certainly real objects. This kind of purity may not be realistic when using datasets with variable image quality, but we expect that it should be fairly standard for images taken by high-quality space telescopes.

`heliostack` will be immensely helpful in elucidating science when applied to other datasets. In particular, it will enable deep searches for solar system objects in surveys with more data, and with sparser temporal cadences, than was previously possible. We plan to apply it to data from HST, JWST (J. A. Stansberry et al. 2021; D. E. Trilling et al. 2025), and eventually Roman. We will also apply it to select ground-based observations of sparsely sampled deep drilling fields such as the DES supernova fields (Dark Energy Survey Collaboration et al. 2016), the DECam Ecliptic Exploration Project (J. A. Stansberry et al. 2024), and eventually LSST. It is also interesting to note that because of the way the algorithm

is constructed, in which the data from different exposures are transformed to a common canvas, it is possible to combine heterogeneous datasets taken by different telescopes, enabling deeper searches than any one telescope can achieve.

In addition to being used as a general search tool, `heliostack` will be useful for several targeted applications in the discovery of solar system objects. One example is the discovery of faint targets for spacecraft such as New Horizons (M. W. Buie et al. 2024) and Lucy (H. F. Levison et al. 2021; L. E. S. Manzano et al. 2025). Another application is doing targeted studies of specific orbital populations, for example CCKBOs (as in this paper), Neptune Trojans, Jupiter Trojans, Planet Nine, or anything else as long as the assumption of two-body motion holds. Since the number of required stacks can be calculated a priori (M. J. Holman et al. 2019), the algorithm has a predictable computational demand, making it possible to decide whether any given study is feasible. Of course, there are still some limitations. For example, complete searches for certain dynamical classes, such as the NEOs, may prove to be computationally infeasible at the moment, but may become possible in the future. It is also not practical to stack images over arbitrarily long time spans; the number of required stacks grows roughly as the time span cubed (G. M. Bernstein et al. 2004; M. J. Holman et al. 2019), so it is best to be discerning when deciding what data are worth stacking.

Finally, we note that while this paper has presented `heliostack` in the context of image data, there is no reason why it must be limited to pixels. Indeed, it is possible to treat both images and catalogs under this same framework; we can think of catalogs as being sparse images, or think of each pixel in an image as being an entry in a catalog. It is possible to use the `heliostack` framework to do trackletless linking in sparse datasets over long time baselines (such as DES and LSST) with well-defined computing requirements. We investigate the application of this technique in forthcoming work (T. R. Ruch et al. 2026, in preparation).

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